A Conditional Low-Szlenk Reduction for the SHAI Problem on $C[0, \omega^\omega]$

arXiv automatic math run `fa_banach_001`

Candidate conditional packet, June 14, 2026

Classification

- **Status:** conditional reduction.
- **Source:** B. Horvath and T. Kania, *Surjective homomorphisms from algebras of operators on long sequence spaces are automatically injective*, arXiv:2007.14112.
- **Target:** Question 3.26 asks whether $C(K)$ has the SHAI property for $K = [0, 1]$, $K = [0, \omega^\omega]$, $K = \beta\mathbb{N} \setminus \mathbb{N}$, and $K = \beta\Gamma$ for an uncountable discrete space Γ.
- **Scope of this packet:** only the subcase $X = C[0, \omega^\omega]$, and only conditional on one operator-ideal equality.

Source Question

The source question appears on PDF page 20, in Question 3.26.

property, but E does not.

Proof. (i) Let $E := \ell_\infty$ and let F be an isomorphic copy of the James space J_2 in E. (Such an F we can always find due to separability of J_2.) Now E has the SHAI property by [12, Proposition 1.2], but F does not have the SHAI property by [12, Example 3.4 (2)].

(ii) Let $E := \ell_1$ and let F be a closed subspace of E such that $E/F \cong J_2$. (Such F exists again by separability of J_2.) Now E has the SHAI property by [12, Proposition 1.2], but E/F does not, as seen above.

(iii) By [15, Theorem 2], there is an uncountable almost disjoint family $\mathcal{A} \subseteq [\mathbb{N}]^\omega$ such that $\mathcal{B}(C_0(K_\mathcal{A}))$ has a character, where $K_\mathcal{A}$ is the Isbell–Mrówka space corresponding to $\mathcal{A}$. Consequently by [12, Lemma 2.2] the Banach space $C_0(K_\mathcal{A})$ does not have the SHAI property. On the one hand, as $\mathfrak{X}_\mathcal{A} \cong C_0(K_\mathcal{A})$, it follows that $\mathfrak{X}_\mathcal{A}$ does not have the SHAI property either. On the other hand $\mathfrak{X}_\mathcal{A}/c_0 \cong c_0(\mathcal{A})$, and it follows from [12, Proposition 1.2] and Theorem A that both c_0 and $c_0(\mathcal{A})$ have the SHAI property. Setting $E := \mathfrak{X}_\mathcal{A}$ and $F := c_0$ concludes the proof. $\square$

3.4. **Open problems.** We conclude this section with some open problems.

As discussed before, $C(K)$-spaces may or may not have the SHAI property. Indeed, the spaces $c_0(\lambda)$, $C(\beta\mathbb{N}) \cong \ell_\infty$ have the SHAI property (Theorem A), whereas $C[0,\omega_1]$, $C_0(K_\mathcal{A})$ and $C(K_0)$ do not have the SHAI property; here K_0 is a Koszmider space without isolated points ([12, Example 2.4 (3) and (6)]). Further naturally arising problems are:

Question 3.26. Does the space $C(K)$ have the SHAI property, where

(i) $K = [0,1]$,
(ii) $K = [0,\omega^\omega]$,
(iii) $K = \beta\mathbb{N} \setminus \mathbb{N}$,
(iv) $K = \beta\Gamma$ for an uncountable discrete space Γ?

Conditional Dependencies

Let

$$X = C[0,\omega^\omega], \qquad J = \overline{\mathscr{G}}_{c_0}(X),$$

where J is the closed ideal of operators on X which are norm limits of operators factoring through c_0. Let

$$\mathscr{SZ}_1(X) = \{T \in \mathscr{B}(X) : \mathrm{Sz}(T) \le \omega\}$$

be the first Szlenk operator ideal restricted to $\mathscr{B}(X)$.

This packet assumes the following unproved dependency.

(D) $J = \mathscr{SZ}_1(X)$.

Equivalently, every operator on $C[0,\omega^\omega]$ with Szlenk index at most ω is a norm limit of operators factoring through c_0. The packet does *not* prove (D).

Proof Intuition

Horvath–Kania already prove that every non-injective SHAI quotient of $\mathscr{B}(C(K))$ kills $\overline{\mathscr{G}}_{c_0}(C(K))$. Thus, for $X = C[0,\omega^\omega]$, the problem is to understand what remains after quotienting by J.

Assumption (D) turns J into the boundary between small and large operators. If $T \notin J$, then $\mathrm{Sz}(T) > \omega$. Bourgain's theorem, combined with Pełczynski's complementation theorem, then forces the identity on X to factor through T. Hence every operator outside J generates the whole algebra $\mathscr{B}(X)$, so J is a maximal ideal. The source kernel containment now says that a non-injective quotient $\mathscr{B}(X) \to \mathscr{B}(Y)$ must have kernel J, making $\mathscr{B}(Y)$ simple. This is impossible for infinite-dimensional Y because of the finite-rank ideal, and finite-dimensional Y is excluded by the square property of $C[0, \omega^\omega]$.

Conditional Theorem

Theorem 1. *Assume (D). Then $C[0, \omega^\omega]$ has the SHAI property.*

Proof. Set $X = C[0, \omega^\omega]$ and $J = \overline{\mathscr{G}}_{c_0}(X)$.

First we show that J is a maximal two-sided ideal of $\mathscr{B}(X)$. Let $T \in \mathscr{B}(X) \setminus J$. By (D), $T \notin \mathscr{S}\mathscr{Z}_1(X)$, so $\mathrm{Sz}(T) > \omega$. The Bourgain theorem quoted in Kania–Laustsen says that if an operator $T : C(K) \to E$ has Szlenk index exceeding ω^α, then T fixes an isomorphic copy of $C[0, \omega^{\omega \cdot \alpha}]$. Taking $\alpha = 1$, $K = [0, \omega^\omega]$, and $E = X$, we find a subspace $W \subseteq X$, $W \cong X$, such that $T|_W$ is bounded below.

The range $T[W]$ is a closed subspace of the separable space X and is isomorphic to $C[0, \omega^\omega]$. By Pełczynski's complementation theorem, $T[W]$ contains a further closed subspace $Z \cong X$ which is complemented in X. Passing to the inverse image of Z inside W, the elementary factorization criterion gives operators $A, B \in \mathscr{B}(X)$ such that

$$ATB = I_X.$$

Thus the two-sided ideal generated by T is all of $\mathscr{B}(X)$. Since this holds for every $T \notin J$, every proper two-sided ideal of $\mathscr{B}(X)$ is contained in J. Hence J is maximal, and $\mathscr{B}(X)/J$ is simple.

Now let $Y \neq 0$ be a Banach space and let

$$\psi : \mathscr{B}(X) \longrightarrow \mathscr{B}(Y)$$

be a surjective algebra homomorphism. Suppose, toward a contradiction, that ψ is not injective. Horvath–Kania Corollary 4.7 gives

$$J = \overline{\mathscr{G}}_{c_0}(X) \subseteq \operatorname{Ker} \psi.$$

The kernel is proper because ψ is surjective onto the nonzero unital algebra $\mathscr{B}(Y)$. By maximality of J, $\operatorname{Ker} \psi = J$. Hence

$$\mathscr{B}(Y) \cong \mathscr{B}(X)/J$$

is simple.

If Y is infinite-dimensional, this is impossible: the finite-rank operators $\mathscr{F}(Y)$ form a nonzero proper two-sided ideal of $\mathscr{B}(Y)$. If Y is finite-dimensional, then $\operatorname{Ker} \psi$ has finite codimension in $\mathscr{B}(X)$. But $X = C[0, \omega^\omega]$ is isomorphic to its square, and Horvath's finite-dimensional reduction lemma for SHAI spaces with a complemented square records that $\mathscr{B}(X)$ has no proper finite-codimensional ideals. This is again impossible.

Therefore no non-injective surjective homomorphism $\mathscr{B}(X) \to \mathscr{B}(Y)$ exists, which is exactly the SHAI property for X. $\square$

Why This Is Conditional

The proof above is complete after (D), but (D) is the central missing operator-ideal statement. The known inclusions and obstructions are:

- Brooker proves that $\mathscr{S}\mathscr{Z}_1$ is a closed operator ideal, and I_{c_0} has Szlenk index ω, so $J \subseteq \mathscr{S}\mathscr{Z}_1(X)$.
- Bourgain plus Pełczynski shows that every operator on $C[0,\omega^\omega]$ outside $\mathscr{S}\mathscr{Z}_1(X)$ generates $\mathscr{B}(X)$.
- What remains unknown here is the reverse inclusion $\mathscr{S}\mathscr{Z}_1(X) \subseteq J$.

Thus any counterexample to (D) would have to be a low-Szlenk operator $T \in \mathscr{B}(C[0,\omega^\omega])$ with $\mathrm{Sz}(T) \leq \omega$ which is not approximately factorable through c_0. Conversely, proving that no such operator exists would promote the $C[0,\omega^\omega]$ subcase from this conditional packet to a genuine partial solution of Question 3.26.

Literature and Duplicate Check

The run indexes were searched for arXiv:2007.14112, the title phrase, SHAI, $C(K)$, $C[0,1]$, $C[0,\omega^\omega]$, $\beta\mathbb{N} \setminus \mathbb{N}$, and related operator-ideal keywords. The only local hit for this exact target before this packet was the lane-12 attempt note.

Focused web and local-source searches on June 14, 2026 found the source paper, Horvath's earlier SHAI paper, Johnson–Phillips–Schechtman's later L^p SHAI paper, Kania–Laustsen's ordinal-operator paper, Brooker's Szlenk-operator ideal paper, and Alspach's paper on operators on $C(\omega^\alpha)$. I found no later paper explicitly answering Question 3.26, and no source proving the equality (D). Kania–Laustsen explicitly state that classifying closed ideals of $\mathscr{B}(C[0,\omega^{\omega^\alpha}])$ for countable $\alpha \geq 1$ appears intractable and is known only in the c_0 case, which is consistent with treating (D) as an unproved dependency rather than a known theorem.

Human Review Recommendation

Reviewers should treat this as a conditional reduction only. The first verification target is the equality

$$\mathscr{S}\mathscr{Z}_1(C[0,\omega^\omega]) = \overline{\mathscr{G}}_{c_0}(C[0,\omega^\omega]).$$

A proof might come from a special factorization theorem for low-Szlenk operators on the first non-c_0 countable ordinal $C(K)$-space. A disproof would likely exhibit a low-Szlenk operator on $C[0,\omega^\omega]$ which does not approximately factor through c_0.

References

[1] B. Horvath and T. Kania, *Surjective homomorphisms from algebras of operators on long sequence spaces are automatically injective*, arXiv:2007.14112.

[2] B. Horvath, *When are full representations of algebras of operators on Banach spaces automatically faithful?*, arXiv:1811.06865.

[3] T. Kania and N. J. Laustsen, *Operators on two Banach spaces of continuous functions on locally compact spaces of ordinals*, arXiv:1304.4951.

[4] P. A. H. Brooker, *Asplund operators and the Szlenk index*, arXiv:1003.5710.

[5] D. E. Alspach, *Operators on $C(\omega^\alpha)$ which do not preserve $C(\omega^\alpha)$*, arXiv:math/9610215.

[6] W. B. Johnson, G. Pisier, and G. Schechtman, *The SHAI property for the operators on L^p*, arXiv:2102.03966.